

Enterprise Zero Trust Architecture & Cryptographic Key Audit

Auditor: CyberDefense Global | **Scope:** Tenant Beta Infrastructure

1. Post-Quantum Cryptographic Migration

All edge ingress gateways and internal service meshes have migrated from RSA-4096 to NIST post-quantum standard algorithms: **ML-KEM (Kyber-768)** for key encapsulation and **ML-DSA (Dilithium-3)** for digital signatures. Mutual TLS 1.3 ephemeral keys are rotated automatically every **48 hours**.

2. SOC2 & Zero Trust Compliance Scorecard

Security Control Domain	Target Policy	Current Compliance	Audit Finding	Risk Level
IAM & MFA Enforcement	FIDO2 WebAuthn	100.0%	Pass (Zero Bypass)	Low
Microsegmentation	Default Deny East-West	98.7%	Pass (1 exception)	Low
SIEM Log Ingestion	Min 100k EPS	165,000 EPS	Pass (Zero drop)	Low
HSM Key Backup	FIPS 140-3 Level 4	100.0%	Pass (Georedundant)	Low

3. Incident Response Threshold

Mean Time to Detect (MTTD) is **42 seconds** and Mean Time to Remediate (MTTR) automated network isolation is **1.8 seconds**.